MSDS 451 – Coding Assignment 2

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1. Problem Description

An investor wants to know how to split their money across several stocks. Each stock has an expected return and a volatility. Stocks also tend to move together, so the investor must think about correlations between them, not just one stock at a time. The goal is to choose the mix that gives the best return for an acceptable level of risk.

This project explores the question in three layers. First, a Monte Carlo simulation generates many random portfolios so we can see the range of outcomes that arbitrary allocations produce. Second, a mean-variance optimizer finds the specific portfolios that give the best return for each level of risk. Third, a stochastic-programming step solves the same optimization using simulated scenarios rather than population parameters, which is closer to how the problem appears in practice when the inputs are estimated from data.

We run the whole pipeline twice. Run 1 uses four hypothetical assets A, B, C, D from the jump-start code, with the parameters given by the professor. Run 2 uses four real stocks - AAPL, MSFT, JNJ, and JPM with parameters calculated from 2015–2024 daily returns.

2. Data

Run 1 — Hypothetical assets

Run 1 uses the parameters provided in the jump-start code: four assets with mean returns of 2%, 7%, 15%, and 20%, and standard deviations of 5%, 12%, 17%, and 25%. Asset A is loosely correlated with the others ($\rho = 0.3$); assets B, C, D are more correlated with each other ($\rho = 0.6$). These values are taken as given and serve mainly to validate that the pipeline reproduces the jump-start study before applying it to real data.

Run 2 — Real stocks

Run 2 uses four well-known stocks: Apple, Microsoft, Johnson & Johnson, and JPMorgan Chase. I downloaded daily adjusted closing prices for 2015 through 2024 using the yfinance package, computed daily simple returns, and annualized them by multiplying the mean by 252 (the number of trading days in a year) and the standard deviation by the square root of 252. The estimated parameters are shown in Tables 1 and 2.

Ticker	Sector	Annual Return	Annual Std. Dev.
AAPL	Technology	27.5%	28.5%
MSFT	Technology	27.3%	27.1%
JNJ	Healthcare	7.6%	18.1%
JPM	Financial	19.9%	27.4%

Table 1. Run 2 input parameters estimated from 2015–2024 daily returns.

	AAPL	MSFT	JNJ	JPM
AAPL	1.000	0.683	0.345	0.416
MSFT	0.683	1.000	0.384	0.441
JNJ	0.345	0.384	1.000	0.400
JPM	0.416	0.441	0.400	1.000

Table 2. Correlation matrix of daily returns.

A few patterns are worth noting before diving into results. AAPL and MSFT have nearly identical returns and volatilities (~27% return, ~28% risk), and they are the most correlated pair (0.68) because both are large-cap technology stocks. JNJ stands out: its return is the lowest by far (about 8%) but so is its volatility (about 18%), making it a more defensive option. JPM sits in the middle on return (20%) but has high volatility (27%), comparable to the tech stocks but with lower expected return.

3. Run 1 — Hypothetical Assets

Run 1 reproduces the jump-start study faithfully. The two-panel figure below shows the long-only opportunity set as a tight blue cluster bounded above by asset D's 20% return, while the shorts-allowed set extends much further in both directions. The black efficient frontier traces the upper boundary of each cloud. This is the visual baseline that motivates the deeper Run 2 analysis: optimization isolates the best portfolios, and short selling expands the range of what is achievable.

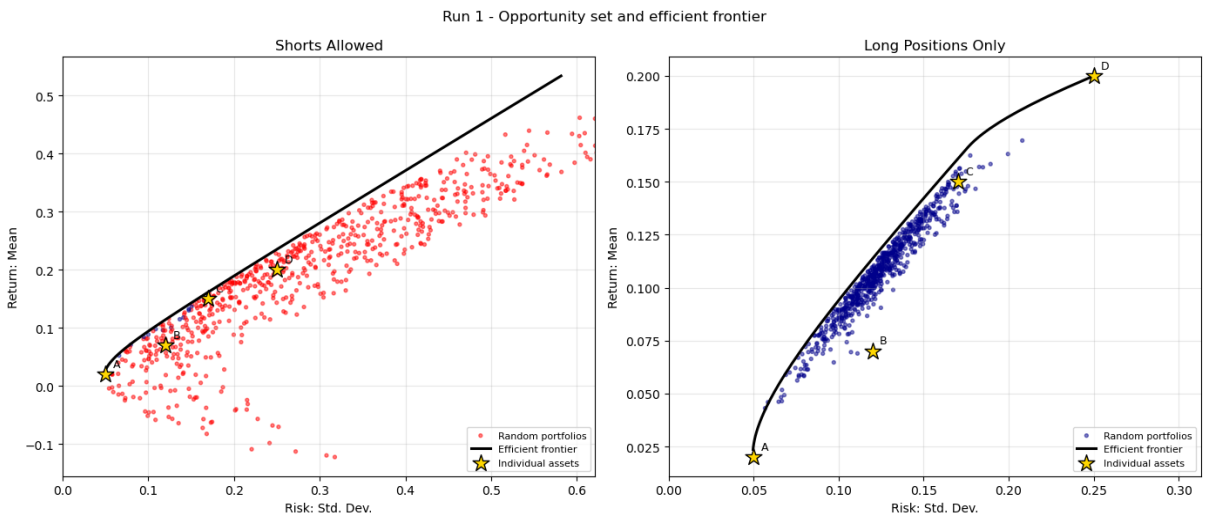

Figure 1.

4. Run 2 — Real Stocks

Figure 2 is the centerpiece of this report. It shows the same kind of analysis as Run 1, but applied to AAPL, MSFT, JNJ, and JPM with parameters estimated from real market history. Read together, the two panels tell almost the entire story of mean-variance portfolio theory.

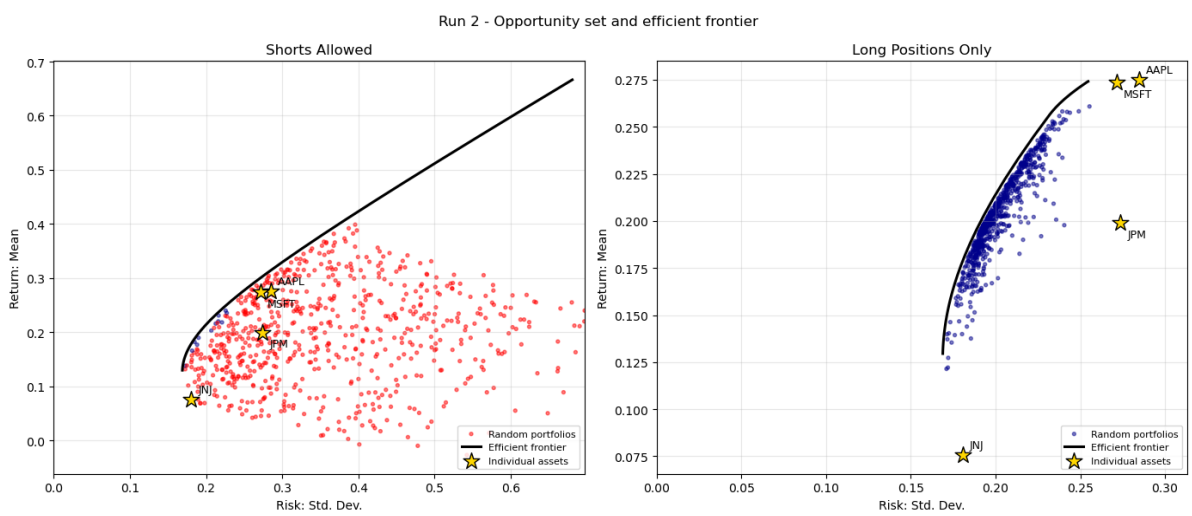

Figure 2

What the long-only panel shows (right-hand side)

The chart illustrates a "long-only" investment scenario, meaning the investor can only buy stocks and not short them. The possible portfolios form a compact cluster, with returns ranging from roughly 12% to 28% and risk levels sitting between 17% and 26%. The efficient frontier—represented by the black line tracing the upper-left edge of this cluster—marks the optimal portfolio for any given level of risk. At the highest risk tolerance, this frontier peaks at MSFT, indicating an all-MSFT portfolio for maximum return. Conversely, at the lowest risk level, the frontier drops down toward JNJ, which becomes the primary holding for maximum stability. The curve between these two extremes represents various blended portfolios of the four stocks.

Notably, both AAPL and JPM fall to the right of the efficient frontier, meaning holding either stock individually is suboptimal compared to holding a diversified mix. AAPL is outperformed because its 27.5% return is only slightly higher than MSFT's 27.3%, yet its volatility is significantly greater. Since the two are highly correlated, adding MSFT to a portfolio reduces risk without sacrificing return. JPM is even more clearly outmatched; at its individual 20% return and 27% risk level, almost any combination of MSFT and JNJ offers better returns with lower volatility. Consequently, the optimization model assigns minimal weight to JPM across the frontier, while AAPL is included only for its slight diversification benefits.

What the shorts-allowed panel shows (left-hand side)

Removing the restriction against short selling drastically alters the investment landscape. The range of possible portfolios explodes outward, with risk levels more than doubling and potential returns stretching beyond 50 percent on the upside while also plunging well below zero on the downside. The line of optimal portfolios climbs significantly higher and reaches a nearly 65 percent expected return. The model achieves this by shorting weaker stocks, specifically JNJ, to aggressively fund a leveraged position in the highest performer, MSFT. Overall, permitting short sales roughly triples the maximum return the model can target. However, this dramatically expanded opportunity set brings massive risks. A randomly chosen portfolio that allows shorting is just as likely to suffer severe losses as it is to earn outsized returns.

Stochastic programming results

The optimization can be done two ways. The deterministic approach plugs the estimated mean and covariance directly into the math. The stochastic-programming approach instead solves the optimization using the 700 simulated return scenarios as the empirical distribution. With normally distributed returns the two methods should give nearly identical answers, with small differences attributable to sampling error in the mean estimate.

Ticker	SAA (shorts)	SAA (long only)	Det. (shorts)	Det. (long only)
AAPL	+0.372	+0.372	+0.340	+0.335
MSFT	+0.417	+0.417	+0.442	+0.434
JNJ	+0.003	+0.003	−0.022	+0.000
JPM	+0.208	+0.208	+0.241	+0.231

Table 3.

All four columns tell the same story: about 40% MSFT, 35–37% AAPL, 20–24% JPM, and very little in JNJ. The optimizer wants the high-return tech names and uses JPM as a moderate diversifier; JNJ's low return makes it unattractive at this risk-aversion level, even though it would help reduce volatility. This intuitively makes sense because MSFT and AAPL have been some of the highest performing large cap stocks over the past decade and have consistently grown in value. The differences across methods are minor. SAA vs. deterministic only shifts weights by a few percentage points, and long-only vs. allowing shorts looks almost identical. At $\lambda = 5.0$, the model just isn't interested in taking on big short positions. Shorting only really shows up when you dial down risk aversion and start acting more aggressively, which is what you see on the far left of the shorts-allowed frontier in Figure 2.

The interpretation is that with this set of four stocks and a moderate risk preference, short selling adds little value. The investor does about as well by simply blending the four long-only. The allocation of stocks is the most significant takeaway with MSFT and AAPL together carry about 80% of the portfolio, JPM rounds it out, and JNJ is essentially absent.

5. Conclusion

The progression from random Monte Carlo allocations to deterministic optimization, and ultimately to stochastic programming, highlights how portfolio optimization can be approached with increasing sophistication. While Monte Carlo simulations map out what is possible, most of those random portfolios are suboptimal for their risk level. Stochastic programming refines this by utilizing actual scenario data rather than abstract population parameters.

Short selling drastically expands potential investment opportunities, but with a significant catch. In a test using real stocks like AAPL, MSFT, JNJ, and JPM, a standard long-only portfolio peaked at roughly a 28% return. While allowing short sales pushed potential returns toward 65%, this only occurred at risk levels most investors would find unacceptable. Consequently, short selling proves much more valuable for aggressive strategies than conservative ones. For the moderately risk-averse investor, the ideal portfolio looks essentially identical whether shorting is allowed or not, typically consisting of about 80% in the two tech stocks with a smaller allocation to JPM.

AI Disclosure

AI was used in the following ways for this assignment. The sample code provided by the professor was originally in R so to convert it to python I used Claude. The Claude code was then ran giving me the Part 1 of the project. I utilized the sample code to construct part 2 of the project. Claude was also used to format tables 1-3 so they were in a more digestible format to include in my write up.